

DA3408 Assignment 2

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Question 1

3) The multi stage image is smaller because the second stage discards the heavy packages required only during the build time like the spam dataset, train.py and the dataset generator.

Question 2

4) Docker Compose allows you to define, configure, and orchestrate multi-container applications—including shared networks, volumes, and startup dependencies—on a single host using a single declarative YAML file, whereas a Dockerfile only builds an image for an isolated, single container. Docker compose is designed for single node development while Kubernetes has a broader scope and orchestrates containerized applications across multi-node production clusters.

Question 3

4) Yes, I would increase parallelism from 4 to 8. With 6 total CPUs available across 3 nodes, the cluster has ample capacity to run all 8 validation pods simultaneously, as 8 pods requesting 500m CPU require only 4.0 CPUs total (2-3 pods per node).

Question 4

4) A Kubernetes Deployment continuously maintains a target replica count through self-healing and zero-downtime rolling updates, ensuring high availability for long-running serving workloads like the Spam API. Conversely, an Indexed Job executes pods to complete a finite, index-bound task and terminates them upon success without restarting completed executions, making it ideal for batch workloads like shard validation. This guarantees web services remain continuously reachable while batch tasks efficiently release cluster compute resources once finished.